Fiscal Year:	FY 2016	**Task Last Updated:**	FY 06/14/2016
PI Name:	Rojas-Pierce, Marcela Ph.D.		
Project Title:	The Role of Vacuole Membrane Fusion in Plant Gravity Perception		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	None		
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City:	Raleigh	**State:**	NC
Zip Code:	27695-7612	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	07/02/2013	**End Date:**	07/01/2017
No. of Post Docs:	2	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA KSC
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Flight Program:			
Flight Assignment:	NOTE: Extended to 7/1/2017 per NSSC information (Ed., 4/5/16)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX13AM49G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The long-term goal of this research is to identify and characterize mechanisms of plant vacuole biogenesis that are critical for gravity perception. A role of the vacuole in controlling gravity perception is well accepted, but the molecular mechanisms by which the vacuole interacts with amyloplasts or regulates their sedimentation during gravitropism are yet to be identified. We recently uncovered a novel vacuole phenotype in agravitropic mutant alleles of VTI11 in Arabidopsis. We predict that this vacuole phenotype of vti11 mutants has negative effects on the organization of actin cables or the cortical endoplasmic reticulum, and therefore prevents amyloplast sedimentation. In addition, we have uncovered a novel regulation of vacuolar membrane fusion by phosphoinositides. Based on our results, we developed the hypothesis that mechanisms of vacuole remodeling that are mediated by phosphoinositide signaling are critical for the perception or transduction of the gravitropic signal in plants. A unique assay for vacuole fusion that we developed		

	will be used to determine the role of specific cellular components in membrane fusion and gravity perception. The specific aims of this research project are to: 1) Determine subcellular phenotypes that impair gravity perception in itt3/vti11. 2) Identify the role of phosphoinositides in gravitropism. 3) Determine the role of the Adaptor Protein (AP-3) complex on vacuole biogenesis and gravitropism. Overall, this research will improve our understanding of vacuole biogenesis, the interactions between the vacuole and other cellular components involved in gravity perception, and the role of phosphoinosidides in gravitropism.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work will contribute to the understanding of the molecular mechanisms that control the response of plants to gravity. In addition, the function of the vacuole in establishing the morphology and dynamics of other organelles will be characterized. These results will enhance the fundamental understanding of how plant cells organize into a defined set of independent organelles.
Task Progress:	We visualized the tonoplast and the ER (endoplasmic reticulum) in endodermal cells using endodermis specific organelle markers, and determined that the association of amyloplasts with vacuoles is abnormal in vti11 alleles. We also used mutants carrying fluorescent markers that label the cytoskeleton and determined that the vacuole is critical for the localization of both actin and microtubule cytoskeletal structures, as well as the ER. In collaboration with the laboratory of Simon Gilroy, we showed that even exposure to a 10 g force cannot displace amyloplasts in itt3/vti11 seedlings. We hypothesized that this is the result of a tight association or tethering of amyloplasts to the vacuolar membranes, and this association is probably indirect. Trans-vacuolar strands (TVS) have been proposed as conduits for amyloplast sedimentation in shoots. Given that itt3/vti11 mutants have very few or no TVS, we wanted to test if this was important for their agravitropic phenotype. Our results indicate that Wortmannin treatment not only promotes vacuole fusion, but has a positive effect on vacuole membrane dynamics as well. These results suggest that Wm-induced vacuole fusion restores the ability of itt3/vti11 to form TVS which could become conduits for amyloplast sedimentation. Jiameng Zheng, a graduate student supported by this grant has continued the characterization of a member of the HOPS complex (for HOmotypic fusion and Protein Sorting) in Arabidopsis. The HOPS complex is a good candidate for a role in gravity sensing because zip2, a mutant allele of the HOPS subunit VPS41, is a suppressor of zig-1 for gravitropism. Jiameng has used reverse genetics and fluorescent protein fusion experiments to identify the functions of HOPS as well as its localization in plant cells. Other experiments that we have carried out identified the affinity of HOPS for phosphoinositides and its ability to bind SNAREs. We have made good progress on this project during the third year of the award. The focus this last year was to finalize all experiments for a publication. We have submitted the following manuscript (under review): Alvarez A, Han SW, Toyota M, Brillada C, Zheng J, Gilroy S, and Rojas-Pierce M. Wortmannin induced vacuole fusion enhances amyloplast dynamics in Arabidopsis zigzag1 hypocotyls and implicates tonoplast in the gravitropic response. Submitted.
Bibliography Type:	Description: (Last Updated: 11/17/2022)
Dissertations and Theses	Zheng J. "Vacuolar SNAREs, HOPS and Phosphoinositides are involved in Homotypic Vacuole Fusion." Dissertation, North Carolina State University, December 2015. , Dec-2015